

Instrumentation

In addition to advanced airframe and engine technology, LSAs often have advanced flight and engine instrumentation. Installation of electronic flight instrumentation systems (EFIS) provides attitude, airspeed, altimeter, vertical speed, direction, moving map, navigation, terrain awareness, traffic, weather, engine data, etc., all on one or two liquid crystal displays. [Figure 17-5] EFIS has become a cost-effective replacement for traditional mechanical gyros and instruments. Compared to mechanical instrumentation systems, EFIS requires almost no maintenance. There are tremendous advantages to EFIS systems as long as the pilot is correctly trained in their use. EFIS systems can cause a “heads down” syndrome and loss of situation awareness if the pilot is not trained to quickly and properly configure, access, program, and interpret the information provided. If EFIS is installed, transition training should include instruction in the use of the specific EFIS in the training airplane. In some cases, EFIS manufacturers or third party products are available for the pilot to practice EFIS operations on a personal computer as opposed to learning their functions in flight.



Figure 17-5. An electronic flight instrumentation system provides attitude, airspeed, altimeter, vertical speed, direction, moving map, navigation, terrain awareness, traffic, weather, and engine data all on one or two liquid crystal displays.

Weather Considerations

Managing weather factors is important for all aircraft but becomes more significant as the weight of the airplane decreases. Smaller, lighter weight airplanes are more easily affected by strong winds (especially crosswinds), turbulence, terrain influences, and other hazardous conditions. [Figures 17-6 and 17-7] LSA Pilots should carefully consider any hazardous weather and effectively use an appropriate set of personal minimums to mitigate flight risk. Some LSAs have a maximum recommend wind velocity regardless of wind direction. [Figure 17-8] While this is not a limitation, it would be prudent to heed any factory recommendations.

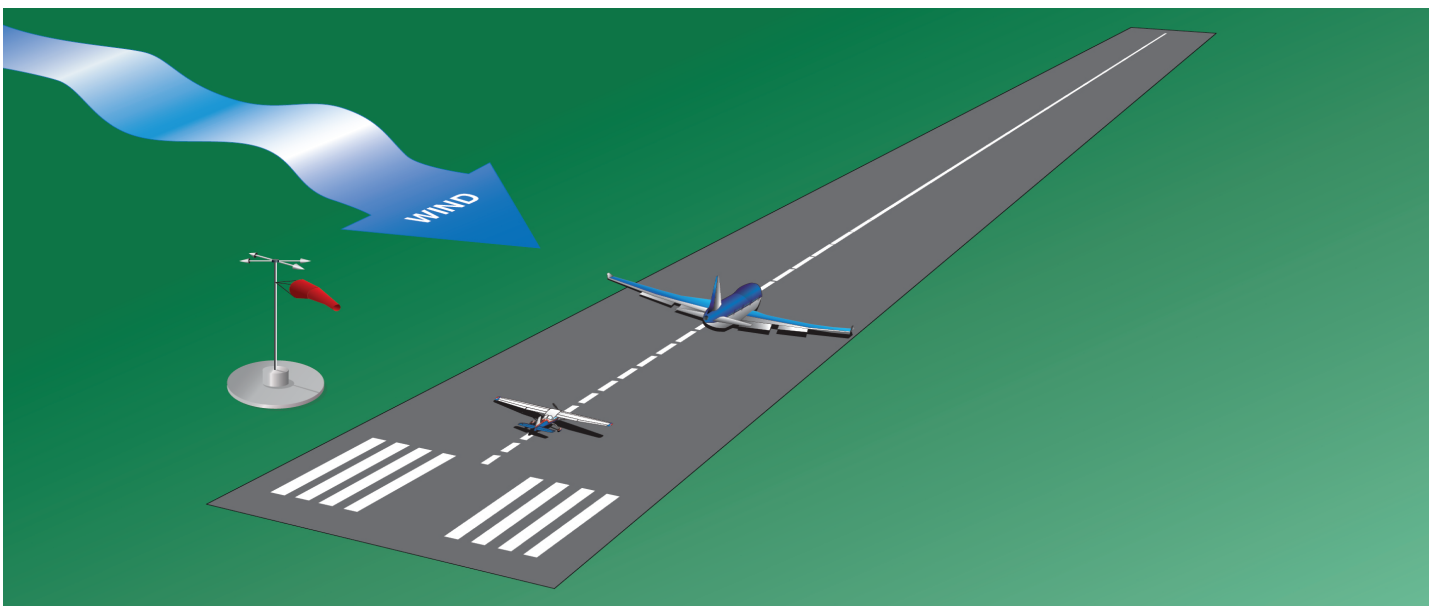


Figure 17-6. Crosswind landing.

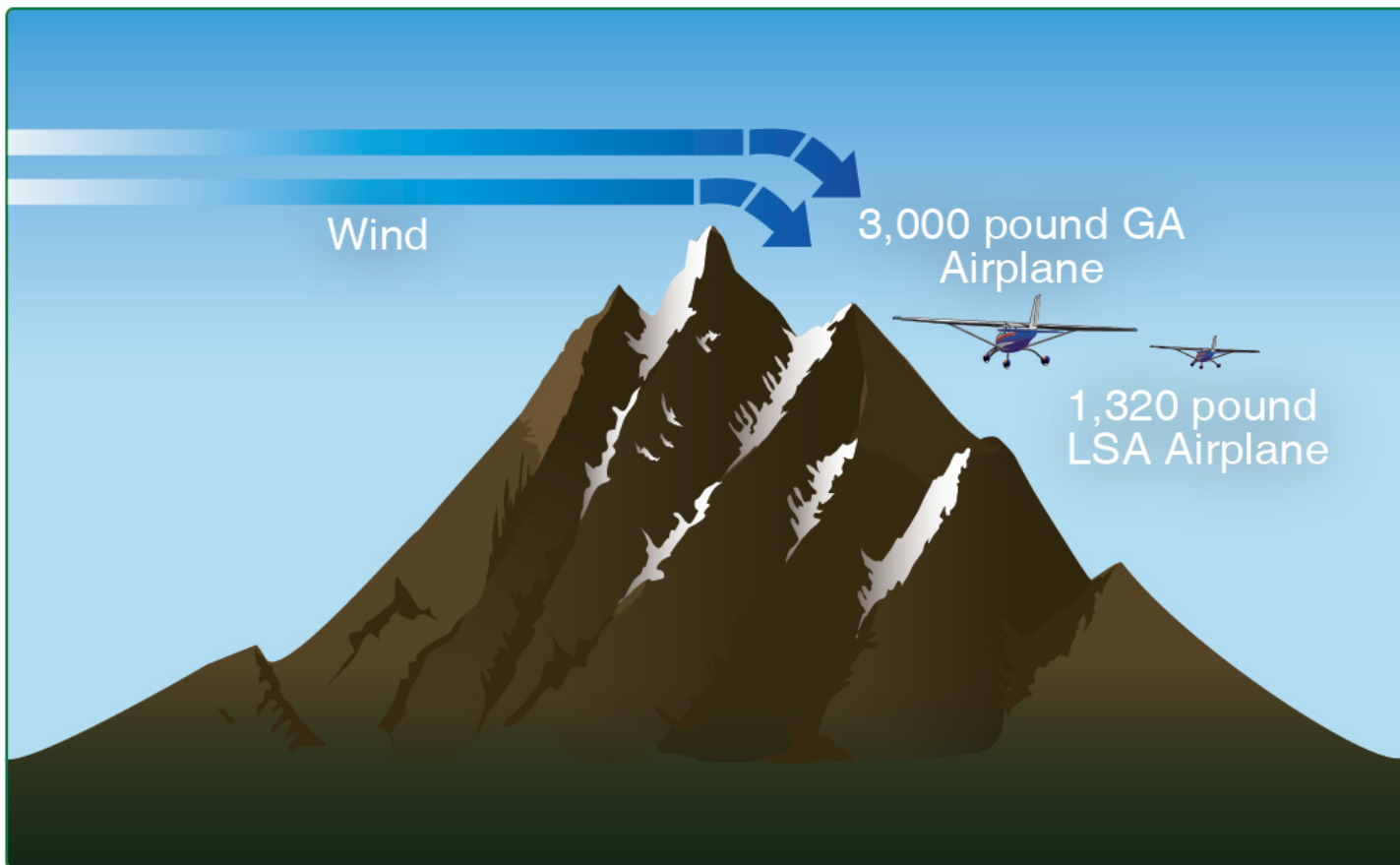


Figure 17-7. Moderate mountain winds can create severe turbulence for LSA.

Maximum Demonstrated Crosswind Velocity

Takeoff or landing.....12 knots

Maximum Recommended Wind Velocity

All operations.....22 knots

Figure 17-8. Example of wind limitations that an LSA may have.

Due to an LSA's lighter weight, even greater distances from convective weather should be considered the norm. While low-level winds that enter and exit a thunderstorm should be avoided by all airplanes, operations in the vicinity of convection should not be attempted in lightweight airplanes. Since it is not always possible to fly in clear, calm air, pilots of lighter weight LSAs should carefully manage all weather-related risks. For example, some consideration should be given to flight activity that crosses varying terrain boundaries, such as grass or water to hard surfaces. Differential heating can cause lighter weight airplanes to experience sinking and lifting to a greater degree than heavier airplanes. Careful planning, knowledge, experience, and an understanding of the flying environment assists in mitigating weather-related risks.

Flight Environment

The skills used to fly LSAs resemble those pilots use when flying any airplane, but the techniques may vary. This section outlines areas that are unique to light-sport airplanes. Most skills learned in a standard airworthiness type certificated airplane are transferable to LSAs; however, since LSAs can vary significantly in performance, equipment, systems, and construction, pilots should seek competent flight instruction and refer to the airplane's POH for detailed and specific information prior to flight.